

Parity-Orthogonal Hardy Channels at the Endpoint Logarithm

Bin Wang

July 2026

Abstract

The endpoint singularity found in RH-312 admits an exact even–odd decomposition. We identify the two closed forms and prove that the Hardy norm is their orthogonal sum. Endpoint convergence is therefore equivalent to convergence in both parity channels separately. This criterion does not assert that either actual noisy channel converges.

1 The two singular channels

Let $L(w) = \log(1 - w) + w = -\sum_{n \geq 2} w^n/n$. Its parity projections are

$$L_{\text{ev}}(w) = \frac{L(w) + L(-w)}{2} = \frac{1}{2} \log(1 - w^2),$$

and

$$L_{\text{odd}}(w) = \frac{L(w) - L(-w)}{2} = w + \frac{1}{2} \log \frac{1 - w}{1 + w}.$$

Both belong to H^2 on the unit disk, and their Taylor supports are disjoint.

Theorem 1.1 (orthogonal endpoint split). *For every $f \in H^2$, define $f_{\text{ev}}(w) = \{f(w) + f(-w)\}/2$ and $f_{\text{odd}}(w) = \{f(w) - f(-w)\}/2$. Then*

$$f = f_{\text{ev}} + f_{\text{odd}}, \quad \|f\|_{H^2}^2 = \|f_{\text{ev}}\|_{H^2}^2 + \|f_{\text{odd}}\|_{H^2}^2.$$

Consequently $f_\sigma \rightarrow 0$ in H^2 if and only if both parity projections tend to zero.

Proof. The even and odd Taylor coefficients occupy disjoint subsets of the orthonormal monomial basis. Parseval gives the identity and the equivalence. $\square$

2 Application to the deterministic endpoint

RH-312 writes the endpoint target as $L + H_{\text{reg}}$ with H_{reg} analytic past the unit circle. Projecting gives

$$(L_{\text{ev}} + H_{\text{reg, ev}}) + (L_{\text{odd}} + H_{\text{reg, odd}}),$$

and both regular pieces retain the larger analytic radius. Thus the two displayed logarithms contain the entire parity-resolved endpoint singularity.

Remark 2.1. The theorem is an exact criterion, not a proof that the noisy modulus complement approximates either channel. No spectral cloud, Gate A–E, or RH conclusion is obtained.